

MTP Midterm Report

Logesh .V (CS24M106)

October 14, 2025

Action Generation Literature Survey

Chapters 1-8

This collection of research papers addresses the fundamental challenge of generating realistic human skeleton animations from motion capture (mocap) data. Despite varying methodological approaches, several common themes emerge across these works. First, there is a consistent focus on **deep generative models**, with multiple papers employing Generative Adversarial Networks (GANs) and their variants (Papers 1, 2, 3, 4, 6), while more recent works explore **diffusion models** (Papers 7, 8) in policy spaces. Second, there is significant emphasis on **temporal modeling**, with architectures ranging from convolutional sequence models (Paper 1) to transformers (Paper 6). Third, many papers grapple with **physical plausibility**, either through explicit physics-based constraints (Papers 2, 7, 8) or implicit learning of physical realism via discriminators. Fourth, **controllable generation** emerges as a key requirement, addressed through action conditioning (Papers 1, 6), task control (Paper 5), and interactive control mechanisms (Papers 2, 3). Finally, the modeling of **skeletal structure** remains central, with several works employing graph convolutional networks (Papers 4, 6) to explicitly capture kinematic relationships between joints.

Chapters 9-12

This collection of papers addresses the challenging problem of inferring 3D human motion capture data from 2D video inputs. Despite their different technical approaches, these works share several fundamental challenges and methodological commonalities. First, they all confront the inherent **2D-to-3D ambiguity problem**, requiring sophisticated methods to lift monocular or sparse multi-view video evidence into accurate 3D skeletal motions (Papers 9, 10, 11, 12). Second, there is consistent emphasis on **temporal coherence and dynamics modeling**, with Papers 9 and 10 explicitly focusing on learning human dynamics from video sequences, while Papers 11 and 12 extend this to longer-term motion generation. Third, these works demonstrate evolving approaches to handling **viewpoint invariance and novel view synthesis**, with Paper 11 specifically addressing sparse multi-view scenarios and Paper 12 (ViMo) representing the state-of-the-art in extracting motions from casual, unconstrained videos. Finally, all methods must bridge the **domain gap** between image-based features and 3D kinematic representations, employing various deep learning architectures to translate visual patterns into geometrically accurate skeletal motions while maintaining physical plausibility across generated sequences.

Contents

Preface	i
1 Convolutional Sequence Generation for Skeleton-Based Action Synthesis (2019)	1
1 Introduction	1
2 Datasets Used	1
3 Methodology	1
3.1 Latent Vector Sequence Generation	1
3.2 Graph Upsampling and Pyramid	2
3.3 Generative Adversarial Training	2
4 Findings and Discussions	3
4.1 Quantitative Results	3
4.2 Qualitative Results	3
4.3 Ablation Studies	3
5 Conclusion	3
5.1 Improvements Done	3
5.2 Improvements Needed	4
2 A GAN-Like Approach for Physics-Based Imitation Learning and Interactive Character Control (2021)	5
1 Introduction	5
2 Dataset Used	5
3 Methodology	5
3.1 Key Improvements Over GAIL	6
3.2 Policy Training	6
4 Findings and Discussions	7
4.1 Imitation Performance	7
4.2 Robustness	7
4.3 Interactive Control	7
4.4 Ablation Studies	7
5 Conclusion	7
5.1 Improvements Made	7
5.2 Limitations and Future Work	8

3	GAN-based Reactive Motion Synthesis with Class-aware Discriminators for Human–Human Interaction (2021)	9
1	Introduction	9
2	Datasets Used	9
3	Methodology	10
3.1	Generator: Part-based Attentive Recurrent Network	10
3.2	Discriminator: Class-aware Dual Classifier	10
3.3	Loss Functions	11
4	Findings and Discussions	11
5	Conclusion	11
5.1	Limitations	12
5.2	Future Directions	12
4	Generative Adversarial Graph Convolutional Networks for Human Action Synthesis (2022)	13
1	Introduction	13
2	Datasets Used	13
3	Methodology	13
3.1	Graph Convolutional Formulation	14
3.2	Conditional Adversarial Training	14
4	Findings and Discussions	15
4.1	Quantitative Results	15
4.2	Ablation Studies	15
5	Conclusion	15
5.1	Improvements Introduced	16
5.2	Limitations and Future Directions	16
5	Composite Motion Learning with Task Control (2023)	17
1	Introduction	17
2	Dataset Used	17
3	Methodology	17
3.1	Core Components	17
3.2	Theoretical Backing	18
4	Findings/Discussions	19
4.1	Key Results	19
4.2	Quantitative Evaluation	19
4.3	Ablation Studies	19
5	Conclusion	19
6	ActFormer: A GAN-based Transformer towards General Action-Conditioned 3D Human Motion Generation (2023)	21
1	Introduction	21
2	Datasets Used	21
3	Methodology	22
3.1	Overview	22
3.2	Single-Person Motion Generation	22
3.3	Multi-Person Motion Generation	22
3.4	Generative Adversarial Training	23

4	Findings and Discussions	23
4.1	Quantitative Results	23
4.2	Ablation Studies	23
4.3	Qualitative Results	23
5	Conclusion	24
5.1	Improvements Done	24
5.2	Improvements Needed	24
7	Policy-Space Diffusion for Physics-Based Character Animation (2025)	25
1	Introduction	25
2	Dataset Used	25
3	Methodology	25
3.1	Theoretical Backing	25
3.2	Mathematical Formulation	26
4	Findings and Discussions	27
4.1	Morphology Adaptation	27
4.2	Terrain Adaptation	27
4.3	Motion Combination	27
4.4	Limitations	27
5	Conclusion	27
5.1	Improvements Made	28
5.2	Needed Improvements	28
8	Policy-space Interpolation for Physics-based Characters (2025)	29
1	Introduction	29
2	Dataset Used	29
3	Methodology	29
3.1	Theoretical Background	29
3.2	Network Architecture and Training	31
4	Findings and Discussions	31
4.1	Interpolation Quality	31
4.2	Failure Cases	31
4.3	Single-Layer Interpolation	31
4.4	Extrapolation	31
4.5	Performance	31
5	Conclusion	32
5.1	Improvements Made	32
5.2	Needed Improvements	32
9	Learning 3D Human Dynamics from Video (2019)	33
1	Datasets Used	33
2	Methodology	33
2.1	Problem Formulation	34
2.2	3D Human Mesh Representation	34
2.3	Temporal Encoder and 3D Regression	34
2.4	Loss Functions	34
2.5	Training Strategy	35
3	Findings and Discussions	35

4	Conclusion	35
4.1	Improvements Made	35
4.2	Improvements Needed	36
10	Predicting 3D Human Dynamics from Video (2019)	37
1	Datasets Used	37
2	Methodology	37
2.1	3D Mesh Representation	37
2.2	Causal Temporal Encoder	38
2.3	Autoregressive Prediction	38
2.4	Loss Functions	38
3	Findings and Discussions	39
3.1	Quantitative Results	39
3.2	Qualitative Results	39
3.3	Failure Modes	39
4	Conclusion	39
4.1	Improvements Done	40
4.2	Future Improvements Needed	40
11	Novel View Synthesis of Human Interactions from Sparse Multi-view Videos (2022)	41
1	Datasets Used	41
2	Methodology	41
2.1	Multi-entity Motion Capture	41
2.2	Layered Neural Scene Representation	42
2.3	Network Training	42
3	Findings / Discussions	43
4	Conclusion	43
4.1	Improvements Done	43
4.2	Future Work & Limitations	43
12	ViMo: Generating Motions from Casual Videos (2024)	44
1	Datasets Used	44
1.1	Primary Training Dataset	44
1.2	Generated Dataset	44
1.3	Other Referenced Datasets	44
2	Methodology	45
2.1	Technical Framework Overview	45
2.2	Diffusion Model Foundation	45
2.3	Network Architecture Details	46
2.4	Motion Representation Details	47
2.5	Loss Functions Formulation	47
2.6	Training and Sampling Procedure	48
2.7	Advanced Applications Implementation	49
2.8	Key Technical Innovations	49
2.9	Applications	50
3	Findings and Discussions	50
3.1	Quantitative Results	50

	3.2	Qualitative Advantages	50
	3.3	Key Insights	51
4		Conclusion	51
	4.1	Improvements Made	51
	4.2	Limitations and Future Work	51

Paper 1

Convolutional Sequence Generation for Skeleton-Based Action Synthesis (2019)

Authors: Sijie Yan, Zhizhong Li, Yuanjun Xiong, Huahan Yan, Dahua Lin

1 Introduction

The paper proposes a **Convolutional Sequence Generation Network (CSGN)** for synthesizing long-duration skeleton-based action sequences. Unlike autoregressive models (e.g., LSTM, GRU), CSGN generates the entire sequence at once by transforming a sequence of latent vectors sampled from a Gaussian Process (GP). The model jointly models spatial and temporal structures using graph convolutions and a novel graph refining scheme.

2 Datasets Used

- **NTU-RGB+D** [?]: A real-world action dataset with 56,000 short clips from 60 action categories, captured by Kinect sensors. Each skeleton has 25 joints.
- **MikuDance**: A new dataset collected by the authors, containing 201 long dancing sequences (3–5 minutes each), totaling 10 hours and 1 million frames. Each skeleton has 46 joints, including finger joints, with rotational information in quaternions.

3 Methodology

3.1 Latent Vector Sequence Generation

The model begins by sampling a sequence of latent vectors $\mathbf{F}_0 = (\mathbf{z}_t)_{t=1}^{T_0}$ from a Gaussian Process:

$$\mathbf{z}_t^{(c)} \sim \mathcal{GP}(0, \kappa), \quad \kappa(t, t') = \exp\left(-\frac{|t - t'|^2}{2\sigma_c^2}\right)$$

where σ_c is the characteristic length-scale for channel c . A spectrum of σ_c values is used to model multiple temporal scales.

Sequence Generation via Graph Convolutions

The generator transforms the latent sequence $\mathbf{F}_0 \in \mathbb{R}^{C_0 \times 1 \times T_0}$ into a skeleton sequence $\mathbf{F} \in \mathbb{R}^{C \times V \times T}$ through a series of blocks:

$$\mathcal{F}_{k+1} = h(\text{Conv}_{st}(\text{Up}_{st}(\mathcal{F}_k)))$$

where:

- Up_{st} : Spatio-temporal upsampling operator
- Conv_{st} : Spatio-temporal graph convolution
- h : Non-linear activation (LeakyReLU)

3.2 Graph Upsampling and Pyramid

A graph pyramid is constructed by iteratively coarsening the skeleton graph. Upsampling is performed by embedding coarser graphs into finer ones and assigning values to new vertices via averaging.

Spatially-Varying Graph Convolution

Unlike standard graph convolution, spatially-varying convolution does *not* share weights across vertices, allowing better modeling of heterogeneous body parts.

3.3 Generative Adversarial Training

The generator G and discriminator D are trained adversarially using WGAN-GP. The discriminator uses:

$$\mathcal{F}_{k+1} = h(\text{Coar}_{st}(\text{Conv}_{st}(\mathcal{F}_k)))$$

where Coar_{st} performs spatio-temporal coarsening.

Action Manipulation via Conditional Sampling

CSGN supports conditional generation (e.g., prediction, completion) via an inverse mapping network H that maps observed skeletons to latent vectors. The conditional sampling process is:

$$p(\mathcal{F}_0 | \mathcal{F}^K) \approx p(\mathcal{F}_0^U | \mathcal{F}_0^K) \cdot p(\mathcal{F}_0^K | \mathcal{F}^K)$$

where $\mathcal{F}^K$ is the observed part and $\mathcal{F}^U$ is the unobserved part.

4 Findings and Discussions

4.1 Quantitative Results

CSGN outperforms autoregressive (ERD, acLSTM) and GAN-based (HP-GAN, Two-Stage) baselines on both NTU-RGB+D and MikuDance datasets in terms of:

- **Short-term generation (FID/IS)**
- **Long-term generation (Ensemble FID/IS, Time FID/IS)**

CSGN achieves lower FID and higher IS scores, indicating better quality and diversity.

4.2 Qualitative Results

- CSGN generates smooth and diverse long sequences (up to 60 seconds).
- Autoregressive models tend to freeze or repeat motions (e.g., ERD).
- CSGN occasionally produces unnatural poses (e.g., twisted bodies, interpenetration).

4.3 Ablation Studies

- **Gaussian Process:** Using a spectrum of length scales (σ_c) improves performance. Extreme values ($\sigma_c = 0$ or ∞) lead to poor generation.
- **Graph Operations:** Gradual upsampling and spatially-varying convolution are critical for high-quality generation.
- **Model Size:** CSGN with halved channels (CSGN-Narrow) still outperforms baselines.

5 Conclusion

CSGN presents a significant advance in skeleton-based action synthesis by leveraging Gaussian Processes and spatio-temporal graph convolutions to generate long, coherent sequences. Its non-autoregressive approach and bidirectional mapping enable flexible manipulation, though challenges remain in physical realism and real-time performance.

5.1 Improvements Done

- Introduced non-autoregressive generation via GP and graph convolutions.
- Proposed a novel graph pyramid for upsampling/downsampling.
- Enabled bidirectional mapping between latent and skeleton spaces for manipulation.
- Collected and released a large-scale long-duration dance dataset (MikuDance).

5.2 Improvements Needed

- Address generation of physically implausible poses.
- Extend to more complex multi-person interactions.
- Improve real-time generation for interactive applications.
- Incorporate semantic or stylistic control beyond conditional sampling.

Paper 2

A GAN-Like Approach for Physics-Based Imitation Learning and Interactive Character Control (2021)

Authors: Pei Xu, Ioannis Karamouzas

1 Introduction

The paper introduces a GAN-inspired framework for training physics-based character controllers without explicit reward engineering, enabling real-time interactive control through policy switching.

2 Dataset Used

The authors use the **LAFAN1 dataset** Harvey2020 for reference motion clips. The dataset contains motion capture recordings at 30Hz from multiple subjects, each 3–5 minutes long, covering a variety of behaviors. Short clips (1–3 seconds) are extracted and retargeted to a simulated humanoid character without manual post-processing.

3 Methodology

The method builds on **Generative Adversarial Imitation Learning (GAIL)** Ho2016, Merel2017, which combines GANs with reinforcement learning. The key idea is to train a policy to imitate expert demonstrations by fooling a discriminator that distinguishes between expert and policy-generated trajectories.

3.1 Key Improvements Over GAIL

To stabilize training and improve performance, the authors introduce three modifications:

1. **Ensemble of Discriminators:** Multiple discriminators share feature extraction layers but have independent final layers. This prevents overfitting and reward vanishing.
2. **Hinge Loss:** Replaces binary cross-entropy with hinge loss:

$$\mathcal{L}_{D_i} = \mathbb{E}[\max(0, 1 + D_i(\mathbf{o}))] + \mathbb{E}[\max(0, 1 - D_i(\tilde{\mathbf{o}}))]$$

where $\mathbf{o}$ is the agent’s observation and $\tilde{\mathbf{o}}$ is the expert’s.

3. **Gradient Penalty:** Enforces 1-Lipschitz continuity via:

$$\mathcal{L}_{D_i}^{GP} = \mathbb{E} \left[(\|\nabla_{\hat{\mathbf{o}}} D_i(\hat{\mathbf{o}})\|_2 - 1)^2 \right]$$

where $\hat{\mathbf{o}}$ is an interpolation between expert and agent samples.

The total discriminator loss for N discriminators is:

$$\mathcal{L}_D = \frac{1}{N} \sum_{i=1}^N (\mathcal{L}_{D_i} + \lambda^{GP} \mathcal{L}_{D_i}^{GP})$$

Observation and State Spaces

- **Discriminator Input:** 5 consecutive frames of local link positions and orientations relative to the root.
- **Policy Input:** 4 consecutive frames including link velocities.
- **Policy Output:** Target joint angles for PD controllers.

3.2 Policy Training

The policy is trained using **Distributed PPO (DPPO)** Heess2017. The reward is the clipped average score from the discriminator ensemble:

$$r_t = \frac{1}{N} \sum_{i=1}^N \text{clip}(D_i(\mathbf{o}_{t-3:t+1}), -1, 1)$$

Interactive Policy Switching

At runtime, a **policy switcher** uses the target policy’s discriminator score to decide when to switch:

$$\frac{1}{N} \sum_{i=1}^N \text{clip}(D_i^{\text{target}}(\mathbf{o}_{t-4:t}), -1, 1) \geq \tau$$

where τ is a threshold. This allows real-time response to user input without motion matching or phase variables.

4 Findings and Discussions

4.1 Imitation Performance

- The method achieves low imitation errors (e.g., 0.06m for walking) without reward tuning.
- Supports both single-clip and multi-clip imitation (e.g., one policy for multiple punch styles).
- Outperforms GAIL baselines and is competitive with state-of-the-art motion tracking methods.

4.2 Robustness

Policies withstand external perturbations (e.g., projectile tests) better than motion-tracking baselines, as they do not rely on fixed reference poses.

4.3 Interactive Control

- Real-time policy switching is feasible using discriminator scores.
- Recovery policies can be auto-activated (e.g., get-up after falling).
- Failure cases include premature switching or infeasible transitions due to pose mismatch.

4.4 Ablation Studies

- Gradient penalty is critical for training stability.
- Velocity in observations harms performance due to increased learning difficulty.
- 32 discriminators strike a balance between overfitting and underfitting.

5 Conclusion

The paper presents a novel GAN-like framework for physics-based imitation learning and interactive character control. By leveraging ensemble discriminators and policy switching, it achieves robust, real-time control without explicit reward engineering or motion tracking. While limitations exist in motion transition quality and long-term memory, the method offers a flexible and efficient alternative for character animation in interactive applications.

5.1 Improvements Made

- Eliminated need for handcrafted reward functions.
- Enabled real-time interactive control without motion tracking.
- Introduced ensemble discriminators, hinge loss, and gradient penalty for stable training.

5.2 Limitations and Future Work

- **Motion Transition Quality:** Transitions can be nonsmooth if reference motions are discontinuous.
- **Unweighted Observations:** All body parts are treated equally; weighting may improve performance.
- **Short-Term Memory:** Policies use only 4-5 frames; long-term dependencies are not captured.
- **Importance Sampling:** Uniform sampling during training is suboptimal; adaptive sampling could help.
- **Goal-Conditioned Learning:** Future work may integrate goal-directed rewards for more responsive control.

Paper 3

GAN-based Reactive Motion Synthesis with Class-aware Discriminators for Human–Human Interaction (2021)

Authors: Qianhui Men, Hubert P. H. Shum, Edmond S. L. Ho, Howard Leung

1 Introduction

The paper introduces a semi-supervised GAN framework for synthesizing reactive human motions in interactive scenarios. The proposed system uses a part-based attentive LSTM generator and a dual discriminator to ensure realistic and class-specific reactive motions. This survey covers the methodology, mathematical foundations, experimental results, and limitations of the work.

The paper addresses the challenge of synthesizing realistic reactive motions in human–human interactions, which is crucial for applications in virtual reality, gaming, and animation. While prior work has focused on single-character motion synthesis, this paper extends GANs to model interactive behaviors between two characters. The key contributions include:

- A semi-supervised GAN system for reactive motion synthesis.
- A part-based attentive recurrent generator that models spatial-temporal dependencies.
- A dual discriminator with binary and multi-class classifiers to enhance motion realism and discriminability.

2 Datasets Used

The authors evaluate their method on three datasets:

- **SBU Kinect Interaction Dataset (SBU)**: Contains 8 interaction types (e.g., *kick*, *push*, *hug*, *shake hands*) with 3D joint positions.
- **Human–Human–Object Interaction (HHOI)**: Includes interactions like *shake hands* and *high-five*, with longer sequences.
- **Character–Character Dataset (2C)**: A high-quality motion capture dataset with *kicking* and *punching* interactions.

3 Methodology

The system takes the active character’s motion as input and synthesizes the reactive character’s motion. The framework includes:

- A **generator** with part-based LSTM encoding and attentive decoding.
- A **discriminator** with binary and multi-class classifiers.
- Auxiliary **loss functions** to enforce physical and temporal constraints.

3.1 Generator: Part-based Attentive Recurrent Network

The generator encodes the active character’s motion using a hierarchical LSTM that processes five body parts (four limbs and trunk) separately. The hidden states for each part are concatenated to form a global spatial representation.

The decoder uses an **attention mechanism** to align the reactive motion with the active motion. The context vector r_t at each decoder step is computed as:

$$r_t = \sum_{s=1}^S \alpha(s, t) h_s,$$

where $\alpha(s, t)$ is the attention weight:

$$\alpha(s, t) = \text{softmax}(V \tanh(W[h_s; \hat{h}_{t-1}])).$$

The LSTM gates in the decoder are updated as:

$$\begin{pmatrix} \hat{i}_t \\ \hat{f}_t \\ \hat{o}_t \\ \hat{u}_t \\ \hat{x}_t \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \\ \tanh \end{pmatrix} \hat{W} \begin{pmatrix} \hat{x}_{t-1}^B \\ \hat{h}_{t-1} \\ r_t \end{pmatrix}.$$

3.2 Discriminator: Class-aware Dual Classifier

The discriminator consists of:

- A **binary classifier** D_b to distinguish real vs. synthetic motions.
- A **multi-class classifier** D_m to predict the interaction type.

The two classifiers share most parameters, making the system efficient. Only the reactive motion is fed to the discriminator to force it to focus on reactive patterns.

3.3 Loss Functions

The total loss is a combination of adversarial and auxiliary losses:

Adversarial Loss

$$\begin{aligned}\mathcal{L}_{sup} &= -\mathbb{E}_{x,y \sim G} \log \frac{p_{D_m}(y|x, y < N+1)}{p_{D_m}(y|x, y = N+1)} + \mathbb{E}_{x,y \sim p_B} \log p_{D_m}(y|x, y < N+1) \\ \mathcal{L}_{unsup} &= \mathbb{E}_{x \sim p_B} \log D_b(x) + \mathbb{E}_{x \sim p_A} \log[1 - D_b(G(x))]\end{aligned}$$

Auxiliary Losses

- **Bone Loss:** Preserves bone lengths:

$$\mathcal{L}_{sll} = \sum_t \sum_j |skl(\hat{x}_t^B, j) - skl_{ref}(j)|$$

- **Continuity Loss:** Ensures smooth motion:

$$\mathcal{L}_{con} = \sum_t \max(\|\hat{x}_{t+\Delta t}^B - \hat{x}_t^B\|^2 - \|\hat{x}_{t+k\Delta t}^B - \hat{x}_t^B\|^2 + \lambda, 0)$$

- **Contractive Loss:** Reduces ambiguity:

$$\mathcal{L}_1 = \sum_t |\hat{x}_t^B - x_t^B|$$

Overall Objective

$$\min_G \max_D \mathcal{L}_{sup} + \mathcal{L}_{unsup} + \alpha \mathcal{L}_{sll} + \beta \mathcal{L}_{con} + \gamma \mathcal{L}_1$$

4 Findings and Discussions

- The model generated realistic and temporally aligned reactive motions across all datasets.
- The multi-class discriminator improved motion quality (Table 1) and enabled diverse reactions.
- The attention mechanism effectively aligned interactions (Fig. 5–6).
- The method outperformed baselines (NN, HMM, DMDP, KRL, ME-IOC) in AFD on most classes (Table 2).
- Ablation studies confirmed the importance of each loss function (Tables 3–4).

5 Conclusion

The paper presents a novel GAN-based framework for reactive motion synthesis in human–human interactions. By combining a part-based attentive generator with a class-aware dual discriminator, the system generates realistic and class-specific reactive motions. Experimental results demonstrate significant improvements over existing methods in both qualitative and quantitative evaluations.

5.1 Limitations

- Difficulty in synthesizing microscopic movements (e.g., hand shaking).
- Dependency on training data quality; artifacts like foot floating occur.
- Inability to model joint orientations for skinned characters.

5.2 Future Directions

- Increase reactive motion diversity via style disentanglement or noise injection.
- Develop online reactive motion synthesis with local temporal attention.
- Incorporate foot contact constraints to reduce floating.
- Use larger datasets and distribution-based metrics (e.g., FID).

Paper 4

Generative Adversarial Graph Convolutional Networks for Human Action Synthesis (2022)

Authors: Bruno Degardin, Joao Neves, Vasco Lopes, etc

1 Introduction

The paper introduces **Kinetic-GAN**, a novel framework that combines Generative Adversarial Networks (GANs) with Graph Convolutional Networks (GCNs) to synthesize realistic and diverse human action sequences from skeleton data. The method addresses key limitations in existing approaches, such as limited action conditioning and poor long-term temporal modeling.

2 Datasets Used

The authors evaluate Kinetic-GAN on three publicly available datasets:

- **NTU RGB+D** – 56,880 videos, 60 action classes, 25 joints per skeleton.
- **NTU-120 RGB+D** – 114,480 videos, 120 action classes, 25 joints.
- **Human3.6M** – 2D human motions, 15 body joints, 10 action classes.

3 Methodology

Kinetic-GAN consists of a generator and a discriminator, both built using **spatiotemporal graph convolutional networks**. The generator synthesizes human action sequences from a latent vector $\mathbf{z}$ and an action class embedding $\mathbf{y}$, while the discriminator distinguishes between real and generated sequences.

3.1 Graph Convolutional Formulation

The skeleton is represented as a spatiotemporal graph $\mathcal{G}_l = (\mathcal{V}_l, \mathcal{E}_l)$ at resolution level l , with N_l joints and T_l frames. The spatial graph convolution is defined as:

$$\mathcal{S}(\mathbf{X}_l) = \sum_{i=1}^p \mathbf{A}_{l_i}^{-\frac{1}{2}} \mathbf{A}_{l_i} \mathbf{A}_{l_i}^{-\frac{1}{2}} \mathbf{X}_l \mathbf{W}_{l_i}$$

where:

- $\mathbf{X}_l \in \mathbb{R}^{N_l \times T_l \times C}$ is the feature map,
- $\mathbf{A}_{l_i}$ is the adjacency matrix for partition i ,
- $\mathbf{W}_{l_i}$ are learnable weights.

Temporal convolution is performed using 1D convolutions across frames.

Generator with Mapping Network

The generator uses a **non-linear mapping network** to transform the latent code $\mathbf{z}$ into an intermediate latent space $\mathcal{W}$, which improves disentanglement and linearity of variation factors. The upsampling process is defined as:

$$\mathbf{X}_{l+1} = \mathcal{T}(\mathcal{S}(Up(\mathbf{X}_l))) + \mathcal{T}(Up(\mathbf{X}_l))$$

where $\mathcal{T}$ and $\mathcal{S}$ are temporal and spatial graph convolutions, and Up is the upsampling operation.

3.2 Conditional Adversarial Training

The model is conditioned on action labels using embedded class representations $\mathbf{y}$. The objective function is based on **WGAN-GP**:

$$\min_G \max_D \mathbb{E}_{x \sim \mathbb{P}_r} [D(x|y)] - \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [D(\tilde{x}|y)] + \lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x}|y)\|_2 - 1)^2]$$

Stochastic Variation via Noise Injection

To increase diversity without affecting skeleton structure, Gaussian noise is injected at each layer:

$$\mathbf{X}_{l+1} = \mathcal{T}(\mathcal{S}(Up(\mathbf{X}_l))) + \mathcal{T}(Up(\mathbf{X}_l)) + \mathbf{r}_{l+1} \mathbf{w}_{l+1}$$

where $\mathbf{r}_{l+1}$ is random noise and $\mathbf{w}_{l+1}$ is a per-channel weight vector.

Truncation Trick

During inference, the latent vector $\mathbf{w}$ is scaled toward the center of $\mathcal{W}$ to improve quality:

$$\mathbf{w}' = \mathbb{E}_{\mathbf{z} \sim \mathbb{P}_{\mathbf{z}}} [f(\mathbf{z})] + \psi (\mathbf{w} - \mathbb{E}_{\mathbf{z} \sim \mathbb{P}_{\mathbf{z}}} [f(\mathbf{z})])$$

where $\psi \leq 1$ controls the trade-off between quality and diversity.

4 Findings and Discussions

4.1 Quantitative Results

Kinetic-GAN outperforms state-of-the-art methods across all datasets and metrics (FID, MMD_a , MMD_s):

Method	FID	MMD_a	MMD_s
c-GAN [30]	27.480	0.919	0.975
Baseline [42]	6.030	0.873	0.954
Kinetic-GAN	3.618	0.772	0.871

Table 4.1: Results on NTU RGB+D (Cross-Subject)

4.2 Ablation Studies

Key findings from ablation studies:

- Removing residual connections degrades performance significantly.
- Regularizing batch normalization reduces spatial artifacts.
- Noise injection improves diversity without structural loss.
- Deeper mapping networks improve disentanglement, especially in cross-view settings.

Action Conditioning

Kinetic-GAN conditions up to **120 actions**, a significant improvement over prior methods (e.g., 10 actions in SA-GCN [44]).

Long-Term Generation

The model generates sequences up to **1024 frames** (34 seconds) without quality degradation, leveraging bidirectional temporal dependencies.

5 Conclusion

Kinetic-GAN represents a significant advancement in human action synthesis by effectively combining GANs and GCNs. It improves sample quality, diversity, and action conditioning while supporting long-term sequence generation. The proposed techniques—mapping networks, noise injection, and graph-based up/downsampling—offer a robust framework for future research in skeleton-based human behavior analysis.

5.1 Improvements Introduced

- Use of GCNs for spatiotemporal modeling of skeleton graphs.
- Mapping network for latent space disentanglement.
- Learnable noise injection for stochastic variation.
- Truncation trick for quality-diversity trade-off.
- Conditional generation of 120 action classes.

5.2 Limitations and Future Directions

- Truncation reduces diversity for $\psi \leq 0.9$.
- Batch normalization removal improves artifacts but complicates conditioning.
- Future work may explore interpretable latent directions, style transfer, and more complex action classes.

Paper 5

Composite Motion Learning with Task Control (2023)

Authors: Pei Xu, Xiumin Shang, Victor Zordan, Ioannis Karamouzas

1 Introduction

The paper proposes a novel framework for physics-based character control that combines composite motion learning and task-driven objectives using deep reinforcement learning.

2 Dataset Used

- **LAFAN1 dataset** (Harvey et al., 2020)
- Other commercial and publicly available motion capture datasets
- Motion data recorded at 30Hz, including:
 - Short clips (1–3 seconds) for single-clip imitation
 - Collections of motions (15–70 seconds) for goal-directed tasks
 - Specific motions: crouch, walk, run, tennis footwork, tennis swing, aiming, juggling

3 Methodology

3.1 Core Components

Full-Body Motion Decoupling

- Character pose represented as $\mathcal{P}_t := \{(\rho_l, q_l, \dot{\rho}_l, \dot{q}_l)\}_{l=1}^{N_{\text{link}}}$
- Body divided into partial-body groups (e.g., upper/lower body)
- Each group i has its own reference motion collection $\mathcal{M}^i$

- Observation space for discriminator: $o_t^i := \{\mathcal{P}_{t-n_i}^i, \dots, \mathcal{P}_t^i, \mathcal{P}_{t+1}^i\}$

GAN-like Imitation Learning

- Multiple discriminator ensembles $D^i = \{D_n^i\}_{n=1}^N$
- Loss function with hinge loss and gradient penalty:

$$\mathcal{L}_{D^i} = \frac{1}{N} \sum_{n=1}^N \left(\mathbb{E}_t[\max(0, 1 + D_n^i(o_t^i))] + \mathbb{E}_t[\max(0, 1 - D_n^i(\tilde{o}_t^i))] + \lambda^{\text{GP}} \mathbb{E}_t[(\|\nabla_{\tilde{o}_t^i} D_n^i(\tilde{o}_t^i)\|_2 - 1)^2] \right)$$

- Imitation reward: $r_t^{D^i} = \frac{1}{N} \sum_{n=1}^N \text{Clip}(D_n^i(o_t^i), -1, 1)$

Multi-Objective Learning Framework

- Policy optimization: $\mathcal{L}_\pi = \sum_{k=1}^K \mathbb{E}_t [\omega_k \bar{A}_t^k \log \pi(\mathbf{a}_t | \mathbf{s}_t, \mathbf{g}_t)]$
- Standardized advantage: $\bar{A}_t^k = \frac{A_t^k - \mathbb{E}_t[A_t^k]}{\sqrt{\text{Var}_t[A_t^k]}}$
- Computationally efficient form:

$$\mathcal{L}_\pi = \mathbb{E}_t \left[\left(\sum_k \omega_k \bar{A}_t^k \right) \log \pi(\mathbf{a}_t | \mathbf{s}_t, \mathbf{g}_t) \right]$$

- **Multi-critic architecture** with **PopArt** normalization for value targets

Incremental Learning

- Cooperative policy: $\pi(\mathbf{a}_t | \mathbf{s}_t, \mathbf{g}_t, \mathbf{a}_t^{\text{meta}}) := \mathcal{N}(\boldsymbol{\mu}_t + \mathbf{w}_t \text{Stop}(\mathbf{a}_t^{\text{meta}}), \boldsymbol{\sigma}_t^2)$
- Weight vector $\mathbf{w}_t$ controls reliance on meta-policy
- Enables sample-efficient training by reusing pre-trained policies

3.2 Theoretical Backing

- **Policy Gradient Theorem:** Foundation for PPO-based optimization
- **Generative Adversarial Networks:** Multiple discriminators enable composite motion learning without manual reward shaping
- **Multi-Objective Optimization:** Avoids scalarization of rewards through independent advantage estimation
- **Linearity of Expectation:** Enables efficient combined policy update (Eq. 7 $\rightarrow$ Eq. 9)

4 Findings/Discussions

4.1 Key Results

- Successfully learned composite motions from multiple reference clips without manual blending
- Achieved physically plausible combinations even with conflicting motions (e.g., punching while running)
- Multi-objective framework balanced imitation and task rewards effectively
- Incremental learning reduced training time from 20+ hours to 30min–2 hours
- Policies generalized to unstructured motion collections for goal-directed tasks

4.2 Quantitative Evaluation

- Imitation error measured using Fast DTW: $e_t = \frac{1}{N_{\text{link}}^i} \sum_{l=1}^{N_{\text{link}}^i} \|p_l - \tilde{p}_l\|$
- Balanced imitation errors across body parts (Table 1)
- Multi-objective approach prevented policy from ignoring difficult motions

4.3 Ablation Studies

- **Without MO:** Failed to balance multiple objectives, preferred easier motions
- **Without PopArt:** Slower convergence, higher variance
- **Without incremental learning:** 10× more samples required for similar performance

5 Conclusion

Contributions

- Novel decoupled full-body control for composite motion learning
- Multi-objective framework with automatic reward balancing
- Sample-efficient incremental learning scheme
- Support for both imitation and goal-directed tasks in unified framework

Highlighted Technical Insights

- **Observation space must satisfy** $\mathbf{o}_t^i \subseteq \{\mathbf{s}_t, \mathbf{s}_{t+1}\}$ for effective policy evaluation
- **Linearity of expectation** enables efficient multi-objective policy update
- **PopArt normalization** stabilizes multi-critic training
- **Dynamic sampling rate** ($\beta \sim \text{Uniform}(0.8, 1.2)$) improves motion combination

Limitations and Future Work

- **Limitations:**

- No phase-locking for multi-phase behaviors
- Cannot distinguish stylistic variations across motion subjects
- Requires retraining for new composite activities
- Body part splitting must be manually defined
- Fails with incompatible motions (e.g., left/right body split)

- **Future Directions:**

- State machines for phase-aware control
- Runtime policy combination without retraining
- Automatic compatibility checking for motion combinations
- More flexible body part groupings

Paper 6

ActFormer: A GAN-based Transformer towards General Action-Conditioned 3D Human Motion Generation (2023)

Authors: Liang Xu, Ziyang Song, etc

1 Introduction

The paper introduces a novel framework for generating 3D human motions conditioned on semantic action labels, supporting both single-person and multi-person interactive actions. The method combines Transformers, GANs, and a Gaussian Process latent prior to achieve state-of-the-art performance.

2 Datasets Used

The paper utilizes several datasets for training and evaluation:

- **NTU-13:** Subset of NTU RGB+D 120 with 13 action categories.
- **NTU RGB+D 120:** Large-scale dataset with 120 action categories (94 single-person, 26 two-person).
- **NTURecon-1P:** SMPL-based version of NTU-1P reconstructed using a motion reconstruction pipeline.
- **BABEL:** Large-scale, high-quality SMPL-based motion dataset with fine-grained action labels.
- **GTA Combat: New synthetic dataset** introduced in this paper, containing complex multi-person combat behaviors (2–5 persons) collected from Grand Theft Auto V.

3 Methodology

3.1 Overview

ActFormer is a GAN-based Transformer framework for action-conditioned 3D human motion generation. It uses:

- A **Gaussian Process (GP) latent prior** to model temporal correlations.
- A **Transformer-based generator** to model spatio-temporal dependencies.
- A **GAN training scheme** for high-quality generative modeling.
- **Alternating Transformer encoders** for multi-person interactions.

3.2 Single-Person Motion Generation

- **Latent Prior:** A latent vector sequence $z \in \mathbb{R}^{T \times C_0}$ is sampled from a Gaussian Process (GP) prior. Each channel is an independent 1D GP with varying length-scales to capture multi-scale temporal correlations.
- **Generator:** The Transformer generator takes z and an action label a . The input tokens are:
 - T tokens from z (after MLP embedding)
 - One class token for action a
- **Positional Encoding:** Learnable positional encodings (PE) are added to tokens.
- **Temporal Transformer (T-Former):** Models temporal correlations across frames.
- **Output:** The class token is discarded; remaining T tokens are projected to motion parameters (root translation l_t and body poses θ_t).

3.3 Multi-Person Motion Generation

- **Input:** The same latent sequence z is shared across P persons to enforce synchronization.
- **Positional Encoding:** 2D PE is used: $PE(t, p) = \text{concat}(TPE(t), PPE(p))$, where TPE is temporal and PPE is person-wise PE.
- **Alternating Transformers:**
 - **I-Former:** Models interactions among persons at each time step.
 - **T-Former:** Models temporal correlations for each person independently.
- **Output:** $P \cdot T$ tokens are projected to (P, T, C) motion sequences.

3.4 Generative Adversarial Training

- **Loss:** Conditional Wasserstein GAN loss:

$$L_D = \mathbb{E}[D(G(z, a), a) - D(\tilde{M}, a)],$$
$$L_G = \mathbb{E}[-D(G(z, a), a)],$$

where $\tilde{M}$ is real motion data.

- **Discriminator:** ST-GCN is used. Multi-person motions are concatenated along the feature dimension. **Permutation augmentation** is applied during training to handle person order invariance.

4 Findings and Discussions

4.1 Quantitative Results

- **Single-Person:** ActFormer outperforms Action2Motion, ACTOR, and CSGN on all datasets (NTU-13, NTURecon-1P, NTU-1P, BABEL) in terms of accuracy and FID scores.
- **Multi-Person:** ActFormer achieves significantly better results on NTU-2P and GTA Combat, especially as the number of persons increases.
- **Adaptability:** ActFormer works well with both SMPL parameters and skeleton joint coordinates, unlike prior methods.

4.2 Ablation Studies

Key findings from ablations:

- **GP Prior:** Replacing GP with Gaussian prior increases FID significantly.
- **Architecture:** Explicit skeleton modeling (GraphConv) is not needed in Transformer-based framework.
- **Positional Encoding:** Learnable PE with 2D combination (temporal + person-wise) works best.
- **Interaction Modeling:** Both shared latent sequence and I-Former are crucial for multi-person generation.
- **Discriminator:** Concatenation + permutation augmentation outperforms symmetric aggregation functions.

4.3 Qualitative Results

- Generated motions are diverse and well-synchronized in multi-person interactions (e.g., cheering, taking photos, combat).
- **Limitation:** ActFormer struggles when persons form separate interaction groups (e.g., in 4P/5P combat scenes).

5 Conclusion

ActFormer represents a significant step towards a general action-conditioned 3D human motion generator. It effectively combines Transformers, GANs, and GP priors to handle diverse motion representations and multi-person interactions. The proposed GTA Combat dataset and extensive experiments demonstrate its superiority over existing methods. Future work should address grouping mechanisms and human-object interactions.

5.1 Improvements Done

- Introduced **GP latent prior** for better temporal modeling.
- Designed a **scalable Transformer architecture** for both single- and multi-person generation.
- Proposed a **novel multi-person interaction modeling** strategy using alternating Transformers.
- Released the **GTA Combat dataset** to facilitate multi-person motion generation research.

5.2 Improvements Needed

- **Grouping Mechanism:** Add explicit modeling of interaction groups for complex multi-person scenarios.
- **Variable Number of Persons:** Current framework requires fixed P ; variable-size groups are not supported.
- **Human-Object Interactions:** Not explored; future work should include object interactions.
- **Training Stability:** GAN training can be unstable; explore diffusion models or other stable training schemes.

Paper 7

Policy-Space Diffusion for Physics-Based Character Animation (2025)

Authors: Michelle Rocca, Sune Darkner, Kenny Erleben, Sheldon Andrews

1 Introduction

The paper introduces a novel framework for generating physics-based character animations using diffusion models in the policy space. The methodology, theoretical foundations, experimental results, and potential improvements are discussed in depth.

2 Dataset Used

The authors used two primary datasets:

- **GroundTruth-set:** Derived from the LaFan1 dataset by Harvey et al. (2020). It consists of motion capture data segmented into shorter clips representing individual motion types.
- **Policy-set:** A custom dataset of control policies trained using imitation learning on the GroundTruth-set. It includes policies for various motions, morphologies, and terrains.

3 Methodology

3.1 Theoretical Backing

The core idea is to treat control policy weights as points on a differentiable manifold. The authors hypothesize that policies for similar motions should lie close in this manifold. This enables generative modeling of policies using diffusion models.

3.2 Mathematical Formulation

Policy Representation

A policy $P_{\mathbf{w}}^{\mathbf{m}}$ with weights $\mathbf{w}$ is trained to imitate a motion clip $\mathbf{m}$. The goal is to ensure that policies for similar motions have similar weights:

$$\mathbf{w}' \approx \mathbf{w}'' \quad \text{if} \quad \mathbf{m}' \sim \mathbf{m}''$$

Common Neighbor Policy (CNP) Regularization

To enforce weight similarity, a regularization term is added during policy training:

$$\mathcal{L} = \mathcal{L}_{\text{AMP}} + \lambda \|\mathbf{w} - \mathbf{w}_{\text{CNP}}\|^2$$

where $\mathbf{w}_{\text{CNP}}$ is a pre-trained policy weight vector, and $\lambda = 0.02$ controls the regularization strength.

Diffusion Model for Policy Generation

A conditional Diffusion Transformer is used to generate policy weights. The model is trained to denoise Gaussian noise into policy parameters. The conditioning is applied by concatenating condition vectors (e.g., morphology parameters) with the input at each diffusion step.

Policy Architecture

The policy network is a compact MLP with:

- Input: 223-dimensional state
- Output: 28-dimensional action (mean of Gaussian distribution)
- Hidden layers: 128 and 32 neurons
- Total parameters: 33k (reduced from 770k in original AMP)

Conditioning Strategy

- Categorical conditions (e.g., motion type) are encoded using PCA-derived continuous vectors.
- Continuous conditions (e.g., height, terrain roughness) are min-max normalized.
- The first 12 PCA components are used, capturing 80% of variance.

Evaluation Metrics

- **Falling Rate (FR):** Percentage of simulations where the character falls.
- **Dynamic Time Warping (DTW):** Measures motion similarity between simulated and ground-truth trajectories.
- **Fréchet Motion Distance (FMD):** Compares distributions of motion features in a latent space.

4 Findings and Discussions

4.1 Morphology Adaptation

- Generated policies adapt to unseen character morphologies without retraining.
- Quantitative results show comparable DTW, FMD, and FR metrics to RL-trained policies.
- The distribution of metrics is narrower for generated policies, indicating more consistent quality.

4.2 Terrain Adaptation

- Policies generated with terrain conditioning improve falling rates by 9.24% on average.
- Combining morphology and terrain adaptation improves performance in 63.73% of cases.

4.3 Motion Combination

- Linear interpolation in policy space produces unrealistic motions.
- Diffusion-generated policies follow curved paths in PCA space, resulting in more realistic motion transitions.
- Combining motions (e.g., Fast-walk with Walk) produces asymmetrical but plausible movements.

4.4 Limitations

- Policies for slow motions (e.g., Old-walk) show poor convergence and clustering.
- Freezing behavior occurs in some motions due to Markovian nature and lack of temporal phase.
- Failures occur with extreme morphologies (e.g., short thighs) or increased character mass.

5 Conclusion

The paper presents a groundbreaking approach to physics-based character animation by leveraging diffusion models in the policy space. The method enables rapid generation of adaptable policies for unseen morphologies, terrains, and motion combinations. While limitations exist, the framework opens new directions for generative motion synthesis and efficient policy representation.

5.1 Improvements Made

- Introduction of CNP regularization for policy similarity.
- Compact policy architecture (33k parameters) without sacrificing motion quality.
- Use of PCA for continuous conditioning of motion types.

5.2 Needed Improvements

- Incorporate simulation feedback during diffusion training to validate motion quality.
- Explore finer-grained motion characteristics and topological analysis of policy space.
- Address freezing behavior in slow motions through alternative policy architectures.
- Expand training set to include extreme morphologies and heavier characters.

Paper 8

Policy-space Interpolation for Physics-based Characters (2025)

Authors: Michele Rocca, Sheldon Andrews, Kenny Erleben

1 Introduction

The work introduces a novel method for blending control policies for physics-based characters by performing linear interpolation directly in the weight space of neural network policies, rather than blending their outputs. This approach reduces computational cost while enabling smooth motion transitions and variation synthesis without retraining. The report details the methodology, theoretical foundations, experimental setup, key findings, and potential improvements.

2 Dataset Used

The paper does not explicitly name a public dataset but relies on **motion capture (Mo-Cap) data** for training imitation policies. The policies are trained using the **Adversarial Motion Prior (AMP)** framework, which uses a discriminator to encourage motion realism. The policies are trained for various skills such as Walk, Run, Crouch, Crawl, Idle, Roll, Get-up, and multiple dance styles.

3 Methodology

3.1 Theoretical Background

The core idea is to interpolate between pre-trained policy networks in **weight space** rather than **action space**. This is motivated by the observation that policies with similar behaviors have similar network weights when trained with a regularization strategy.

Policy Regularization

The authors extend the **Common Neighbor Policy (CNP)** regularization strategy to a **graph-based structure**. The loss function for training a policy is:

$$\mathcal{L} = c_{\text{Disc}}\mathcal{L}_{\text{Disc}} + c_{\text{Task}}\mathcal{L}_{\text{Task}} + \gamma\|\mathbf{w} - \mathbf{w}_{\text{Reg}}\|^2$$

where:

- $\mathbf{w}_{\text{Reg}}$: weights of a pre-trained policy used as a regularizer
- $c_{\text{Disc}}, c_{\text{Task}}$: weights for imitation and task losses
- γ : regularization strength

Graph-Based Regularization

Policies are organized in a graph where edges represent regularization directions (see Fig. 2 in the paper). For example:

- Walk $\rightarrow$ Sneaky $\rightarrow$ Jogging $\rightarrow$ Running
- Idle $\rightarrow$ Dance1, Dance2, ...

Linear Interpolation in Weight Space

Given a path between policies π_0 and π_n via intermediate policies $\pi_1, \dots, \pi_{n-1}$, the interpolated policy at parameter $s \in [0, 1]$ is:

$$\pi(s) = (1 - s')\pi_k + s'\pi_{k+1}$$

where $s' = ns - k$ for $k = \lfloor ns \rfloor$.

For multiple policies, barycentric coordinates are used:

$$\pi = \lambda_1\pi_1 + \lambda_2\pi_2 + \dots + \lambda_m\pi_m$$

with $\lambda_i \geq 0$ and $\sum \lambda_i = 1$.

Sampling Strategy

To test interpolation, coefficients λ_i are sampled with rejection to ensure:

- One policy has $\lambda > 0.80$, or
- Two policies have $\lambda > 0.40$, or
- Three policies have $\lambda > 0.25$

Evaluation Metrics

- **Fréchet Motion Distance (FMD)**: Measures motion similarity using latent space encodings of pose sequences.
- **Success/Failure Criteria**: Based on motion type (e.g., falling for standing motions, distance traveled for crawling).

3.2 Network Architecture and Training

- **Policy Network:** MLP with two hidden layers [128, 32], ReLU activations.
- **Input:** 225-D state (root height, joint positions, velocities, target location).
- **Output:** 28-D action vector (mean of Gaussian).
- **Training Phases:**
 1. Imitation-only ($c_{\text{Disc}} = 1$)
 2. Task learning ($c_{\text{Task}} > 0$)
 3. Fine-tuning on rough terrain

4 Findings and Discussions

4.1 Interpolation Quality

- Weight-space interpolation produces motions **visually indistinguishable** from action interpolation.
- **Computational cost** is reduced from $O(cnp)$ to $O(cp)$, where c = number of characters, n = number of policies, p = parameters.
- Interpolation works best when policies have **similar state distributions** and are **compatible** (e.g., Walk, Crouch, Run).

4.2 Failure Cases

- **Get-up** and **Roll** are not suitable for interpolation due to out-of-distribution states.
- **Crawling** often fails when mixed with upright policies.
- **Dance policies** show mixed success, with failures attributed to incompatible motion styles.

4.3 Single-Layer Interpolation

Interpolating only specific layers (e.g., hidden or output) produces motion variations that are **not achievable via action interpolation**, revealing the expressive power of weight manipulation.

4.4 Extrapolation

Small extrapolation ($\sim 10\%$) sometimes works, but larger values lead to invalid policies.

4.5 Performance

- Precomputed interpolation is efficient.
- Real-time interpolation adds ~ 1.2 ms per step.
- Overall speedup is limited by non-inference overhead (e.g., state computation).

5 Conclusion

Rocca et al. present a promising approach for efficient and expressive motion synthesis via weight-space interpolation. The method is particularly beneficial for crowd simulation and real-time interactive applications. While it requires careful regularization and compatibility between policies, it opens new avenues for understanding and manipulating neural motion representations.

5.1 Improvements Made

- Introduced **graph-based regularization** to structure the weight space.
- Enabled **real-time policy switching** via weight interpolation.
- Showed that **single-layer interpolation** can produce novel motions.

5.2 Needed Improvements

- **Better regularization graphs** for dissimilar motions.
- **Reward engineering** for challenging skills like jumping.
- **Dimensionality reduction** of policy manifolds for interpretability.
- **Integration with language-conditioned controllers.**
- **Optimization of observation computation** to realize full speedup potential.

Paper 9

Learning 3D Human Dynamics from Video (2019)

Authors: Angjoo Kanazawa, Jason Y. Zhang, Panna Felsen, Jitendra Malik

1 Datasets Used

The paper utilizes a multi-tiered dataset strategy to train and evaluate the proposed model, HMMR (Human Mesh and Motion Recovery):

- **Human3.6M:** Large-scale motion capture dataset with 3D ground truth annotations. Used for training and validation.
- **Penn Action:** In-the-wild video dataset with 2D pose annotations for sports actions.
- **NBA (Ours):** Custom dataset of basketball players shooting 3-pointers, with 2D pose annotations.
- **VLOG-people:** Subset of the VLOG lifestyle dataset, filtered for consistent OpenPose detections. Provides pseudo-ground truth 2D poses.
- **InstaVariety:** Large-scale dataset collected from Instagram using activity-specific hashtags. Provides pseudo-ground truth 2D poses via OpenPose.
- **3DPW (3D Poses in the Wild):** Used exclusively for testing; contains in-the-wild videos with 3D ground truth from IMUs.

2 Methodology

The proposed framework, HMMR, learns a temporal representation of 3D human dynamics from video and transfers it to static images. The methodology consists of the following components:

2.1 Problem Formulation

Given a video $V = \{I_t\}_{t=1}^T$, the goal is to learn a temporal encoder f_{movie} that maps a window of image features $\{\phi_t\}$ to a dynamics-aware representation Φ_t . A hallucinator h is also learned to map a single image feature ϕ_t to $\tilde{\Phi}_t$, enabling motion prediction from static images.

2.2 3D Human Mesh Representation

The 3D human body is represented using the SMPL model:

$$\mathcal{M}(\beta, \theta) \in \mathbb{R}^{N \times 3}, \quad N = 6890$$

where:

- $\beta \in \mathbb{R}^{10}$: shape parameters
- $\theta \in \mathbb{R}^{72}$: pose parameters (global rotation + 23 joints)
- $\Pi = [s, t_x, t_y]$: weak-perspective camera parameters

3D joints are regressed as $X = W\mathcal{M}(\beta, \theta)$, and projected to 2D as $x = \Pi(X)$.

2.3 Temporal Encoder and 3D Regression

The temporal encoder f_{movie} is a 1D fully convolutional network (ResNet-based) that processes a window of 13 frames. It outputs Φ_t , which is fed into:

- A 3D regressor f_{3D} : Predicts $\Theta_t = \{\beta, \theta, \Pi\}$
- Delta predictors $f_{\Delta t}$: Predict pose changes $\Delta\theta$ for $t \pm \Delta t$

2.4 Loss Functions

The model is trained with a combination of losses:

$$\begin{aligned} L_{2D} &= \|v_t(x_t - \hat{x}_t)\|_2^2 \\ L_{3D} &= \|\Theta_t - \hat{\Theta}_t\|_2^2 \quad (\text{if available}) \\ L_{\text{adv prior}} &= \sum_k (D_k(\Theta) - 1)^2 \\ L_{\beta \text{ prior}} &= \text{Shape prior} \\ L_{\text{const shape}} &= \sum_{t=1}^{T-1} \|\beta_t - \beta_{t+1}\| \\ L_{\text{hal}} &= \|\Phi_t - \tilde{\Phi}_t\|_2 \end{aligned}$$

The overall objective is:

$$L = L_{\text{temporal}} + L_{\text{hal}} + L_t(\tilde{\Phi}_t) + \sum_{\Delta t} L_{t+\Delta t}(\tilde{\Phi}_t)$$

2.5 Training Strategy

The model is trained in a semi-supervised manner using:

- Fully supervised data (Human3.6M)
- 2D-annotated in-the-wild data (Penn Action, NBA)
- Pseudo-ground truth 2D poses from unlabeled videos (VLOG, InstaVariety)

3 Findings and Discussions

- **Temporal Context Improves Smoothness:** Using temporal context reduces acceleration error by 56% compared to single-view methods.
- **Dynamics Prediction Helps:** Auxiliary loss for predicting pose changes improves 3D and 2D accuracy.
- **Pseudo-Ground Truth Boosts Performance:** Training with pseudo-ground truth 2D poses from unlabeled videos (e.g., InstaVariety) reduces 3D error by 9% and increases 2D accuracy by 8%.
- **State-of-the-Art on 3DPW:** HMMR achieves SOTA results on 3DPW without fine-tuning.
- **Hallucinated Dynamics:** The model can predict plausible past and future motion from a single image, though performance is limited for ambiguous motions.

4 Conclusion

HMMR presents a unified framework for learning 3D human dynamics from video and static images. By leveraging temporal context and pseudo-ground truth from unlabeled videos, it achieves state-of-the-art performance in 3D pose estimation and motion prediction. The work demonstrates the potential of semi-supervised learning for 3D human understanding in unconstrained environments.

4.1 Improvements Made

- Introduction of a temporal encoder for smooth 3D mesh prediction from video.
- A hallucinator for predicting 3D dynamics from static images.
- Effective use of pseudo-ground truth 2D poses from unlabeled Internet videos.
- Joint training across multiple datasets with varying levels of supervision.

4.2 Improvements Needed

- Better handling of occlusions and multi-person interactions.
- Improved robustness to rare or extreme poses.
- Addressing ambiguities in motion prediction from static images.
- Extending to longer-term motion forecasting.

Paper 10

Predicting 3D Human Dynamics from Video (2019)

Authors: Jason Y. Zhang, Panna Felsen, Angjoo Kanazawa, Jitendra Malik

1 Datasets Used

The authors used the following datasets for training and evaluation:

- **Human3.6M** [?]: A large-scale motion capture dataset with 3D ground truth annotations. Used Subjects 1, 6, 7, 8 for training; Subject 5 for validation; and Subjects 9 and 11 for testing.
- **Penn Action** [?]: In-the-wild sports videos with 2D keypoint annotations for 15 activities (e.g., bowling, golf swing).
- **NBA** [?]: Videos of professional basketball players attempting 3-point shots, with 2D keypoint annotations.
- **InstaVariety** [?]: Large-scale Internet videos from Instagram with pseudo-ground truth 2D keypoints from OpenPose [?].

2 Methodology

The paper introduces **Predicting Human Dynamics (PHD)**, an autoregressive framework for predicting future 3D human mesh sequences from past video frames. The approach consists of three main components:

2.1 3D Mesh Representation

The human body is represented using the **SMPL model** [?], a differentiable function $\mathcal{M}(\beta, \theta)$ that outputs a 3D mesh with 6890 vertices. The parameters are:

- Pose parameters $\theta \in \mathbb{R}^{72}$ (axis-angle representation for 23 joints + global rotation)

- Shape parameters $\beta \in \mathbb{R}^{10}$ (PCA coefficients)

A weak-perspective camera model $\Pi = [s, t_x, t_y]$ is used for 2D projection.

2.2 Causal Temporal Encoder

A causal convolutional network f_{movie} encodes past image features ϕ_t (from ResNet-50) into a latent "movie strip" Φ_t representing motion history:

$$\Phi_t = f_{\text{movie}}(\phi_{t-r}, \dots, \phi_t)$$

A 3D regressor f_{3D} maps Φ_t to SMPL parameters $\Theta_t = [\theta_t, \beta_t]$ and camera Π_t :

$$\Theta_t, \Pi_t = f_{3D}(\Phi_t)$$

2.3 Autoregressive Prediction

An autoregressive model f_{AR} predicts future latent representations:

$$\tilde{\Phi}_t = f_{AR}(\Phi_1, \dots, \Phi_{t-1})$$

The joint distribution of future sequences is modeled as:

$$p(\Phi) = \prod_{t=1}^T p(\Phi_t | \Phi_1, \dots, \Phi_{t-1})$$

The predicted meshes are decoded via $f_{3D}(\tilde{\Phi}_t)$.

2.4 Loss Functions

The model is trained with a combination of losses:

- **3D Loss:** $L_{3D} = \|X_t - \tilde{X}_t\|_2^2 + \|\theta_t - \bar{\theta}_t\|_2^2 + \|\beta - \hat{\beta}\|_2^2$
- **2D Reprojection Loss:** $L_{2D} = \|v_t \odot (x_t - \hat{x}_t)\|_2^2$
- **Adversarial Prior Loss:** $L_{\text{adv prior}}$ from [?]
- **Shape Prior:** $L_{\text{beta loss}}$ from [?]
- **Consistency Loss:** $L_{\text{const}} = \sum_{t=1}^{T-1} \|\beta_t - \beta_{t+1}\|_2^2$
- **Distillation Loss:** $L_{\text{movie strip}} = \|\Phi_t - \tilde{\Phi}_t\|$

Total loss for autoregressive training:

$$L_{AR} = \sum_t L_t + L_{\text{const}} + \sum_t L_{\text{movie strip}}$$

3 Findings and Discussions

3.1 Quantitative Results

- **Latent vs. Pose Space:** Autoregressive prediction in latent space (AR on Φ) outperforms direct pose prediction (AR on Θ), especially for long-term predictions (Table 2).
- **Baseline Comparison:** PHD significantly outperforms no-motion (Constant) and nearest-neighbor (NN) baselines.
- **Single-Image vs. Sequence:** Outperforms single-image future prediction methods [?, ?] on Penn Action (Table 3).
- **Dynamic Time Warping (DTW):** Using DTW for evaluation accounts for timing discrepancies and shows more robust performance (Tables 4-5).

3.2 Qualitative Results

- The model captures both periodic (walking) and complex sports motions (bowling, tennis).
- Latent space predictions preserve motion type better than pose space predictions (Figure 4).
- Discovered "statue moments" — frames that maximize prediction improvement — align with expressive poses (Figure 5).

3.3 Failure Modes

- **Ambiguity:** 2D keypoints can have multiple 3D interpretations (Figure 13).
- **Poor Conditioning:** Unusual viewpoints degrade performance (Figure 14).
- **Little Motion:** Difficulty predicting start of actions from static preparation (Figure 15).
- **Drifting:** Predictions degrade beyond 35 frames due to limited training sequence length (Figure 16).

4 Conclusion

The paper presents a novel autoregressive framework for 3D human motion prediction from video, demonstrating strong performance on both controlled and in-the-wild datasets. By learning a causal latent representation and leveraging weak supervision, the approach enables practical applications in human-robot interaction and autonomous systems.

4.1 Improvements Done

- First method to predict 3D human mesh sequences from video (not 3D input).
- Causal temporal encoding enables autoregressive future prediction.
- Weakly-supervised training allows use of in-the-wild videos without 3D labels.
- Curriculum learning mitigates drifting during autoregressive generation.

4.2 Future Improvements Needed

- **Multimodality:** Modeling multiple possible futures (e.g., speed, motion type).
- **Occlusion Handling:** Robustness to significant occlusions.
- **Environmental Constraints:** Incorporating scene affordances and physics.
- **Longer Sequences:** Training on longer video tracks to reduce drifting.

Paper 11

Novel View Synthesis of Human Interactions from Sparse Multi-view Videos (2022)

Authors: Qing Shuai, Chen Geng, Qi Fang, Sida Peng, Wenhao Shen, Xiaowei Zhou, Hujun Bao

1 Datasets Used

- **Multi-Human Dataset:** A custom dataset with 4 dynamic scenes involving human interactions, captured by 22 synchronized cameras. Eight cameras were used for training and four for testing.
- **ST-NeRF Dataset:** Provided by Zhang et al. (2021), containing sequences with 16 cameras covering up to 180 degrees.
- **Soccer Dataset:** A real-world soccer game captured using 8 static GoPro cameras at 60 fps, used to evaluate in-the-wild performance.

2 Methodology

The proposed system reconstructs a layered neural representation of dynamic multi-person scenes from sparse multi-view videos. The pipeline consists of three main stages:

2.1 Multi-entity Motion Capture

- **Camera Calibration & Synchronization:** Cameras are calibrated and synchronized manually. A pose-guided synchronization method is introduced to compensate for temporal misalignment by minimizing 2D reprojection loss:

$$\min_{[\Delta t_v]} \sum_{v=1}^V \mathcal{L}_{2d}(\theta + \Delta t_v \hat{\theta}, \beta, R + \Delta t_v \hat{R}, T + \Delta t_v \hat{T}, W_v, w_v)$$

- **Human Pose Detection & Matching:** 2D human poses and object bounding boxes are detected and matched across views. 3D keypoints are triangulated.
- **Parametric Model Fitting:** SMPL+H models are fitted to 3D keypoints using the loss:

$$\mathcal{L}_{\text{mocap}} = \mathcal{L}_{3\text{d}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_t \mathcal{L}_t$$

where $\mathcal{L}_{3\text{d}}$ is the 3D keypoint distance, $\mathcal{L}_{\text{reg}}$ regularizes SMPL parameters, and $\mathcal{L}_t$ enforces temporal smoothness.

2.2 Layered Neural Scene Representation

The scene is represented as multiple neural radiance fields (NeRF) for each human, object, and background.

- **Human Representation:** Uses NeuralBody’s approach with latent codes anchored on SMPL vertices.
- **Object Representation:** Balls are modeled as radiance fields with constant densities and time-varying colors. The transformation to canonical space is:

$$o^m(p_{tt}), c^m(p_{tt}, d) = \delta^m(p_{tt} - \mathbf{T}_t^m), \epsilon^m(p_{tt} - \mathbf{T}_t^m, d, \mathbf{l}_t)$$

- **Background Representation:** A NeRF with temporal latent codes, limited to a region around the ground plane.
- **Rendering:** Layered volume rendering is performed. The color of a ray $\mathbf{r}$ is computed as:

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^N o(p_i) \prod_{j < i} (1 - o(p_j)) c(p_i, \mathbf{r})$$

2.3 Network Training

The model is trained with:

- **Rendering Loss:**

$$\mathcal{L}_{\text{rgb}} = \sum_{\mathbf{r} \in \mathcal{R}} \|\hat{C}(\mathbf{r}) - C(\mathbf{r})\|_2^2$$

- **Sparsity Regularization:** An entropy loss encourages each ray to be assigned to one layer:

$$\mathcal{L}_{\text{ent}} = - \sum_{\mathbf{r} \in \mathcal{R}} \sum_{m=1}^{|\alpha(\mathbf{r})|} \alpha_m \log(\alpha_m)$$

- **Keypoint Supervision:** Uses detected 2D keypoints to guide layer assignment:

$$\mathcal{L}_{\text{kpt}} = - \sum_{\mathbf{r} \in \mathcal{R}} w(\mathbf{r}) \|\alpha(\mathbf{r}) - \hat{\alpha}(\mathbf{r})\|$$

3 Findings / Discussions

- The proposed method outperforms NeuralBody and ST-NeRF in both synthetic and real-world datasets, especially in crowded scenes with close interactions.
- The weakly-supervised layer decomposition eliminates the need for precomputed instance masks, which are often inaccurate.
- Pose-guided synchronization significantly reduces artifacts caused by temporal misalignment.
- The method achieves high-quality novel view synthesis and accurate instance segmentation even with only 8 input views.
- On the ST-NeRF dataset, the method achieves comparable results with half the number of cameras and superior performance with the same number.

4 Conclusion

The paper presents a novel system for synthesizing free-viewpoint videos of multiple humans from sparse multi-view videos. By introducing a weakly-supervised layered neural representation and addressing practical challenges like camera synchronization, the method achieves state-of-the-art results in both synthetic and real-world scenarios. The work opens up possibilities for editable free-viewpoint video production in challenging, in-the-wild settings.

4.1 Improvements Done

- Weakly-supervised layer decomposition without requiring instance masks.
- Pose-guided synchronization for handling asynchronous cameras.
- Layered neural representation for multiple humans and objects.
- Application to in-the-wild scenes using sparse GoPro cameras.

4.2 Future Work & Limitations

- Currently limited to static cameras and simple objects (e.g., balls).
- Does not handle general object categories or complex backgrounds.
- Future directions:
 - Extend to moving cameras or monocular video.
 - Incorporate 6DoF object pose tracking for general objects.
 - Use offline background scanning for improved background rendering.

Paper 12

ViMo: Generating Motions from Casual Videos (2024)

Authors: Liangdong Qiu, Chengxing Yu, Yanran Li, Zhao Wang, Haibin Huang, Chongyang Ma, Di Zhang, Pengfei Wan, Xiaoguang Han

1 Datasets Used

The paper utilizes several datasets for training and evaluation:

1.1 Primary Training Dataset

- **AIST++:** Contains 1,408 high-quality dance motions with music from 10 genres, with 3D motions and 2D videos from 8 camera angles.
- **Training Split:** 8 dance genres for training, **break** and **middle hip-hop** for testing.

1.2 Generated Dataset

- **Chinese Classic Dance Dataset:**
 - 52 solo-dance videos from Chinese annual dancing competition
 - Processed with AlphaPose for 2D pose extraction
 - Generated 750 3D motion clips (63 minutes total) using ViMo
 - Used for music-to-motion generation and style transfer tasks

1.3 Other Referenced Datasets

- Human3.6M
- AMASS
- KIT Motion-Language
- Finedance

2 Methodology

ViMo addresses the novel problem of **video-to-motion generation** from *casual videos* - videos characterized by complex camera movements, montages, occlusions, and varying perspectives. Given an input video $\mathcal{V}$ containing a single person performing actions, the goal is to generate multiple plausible 3D human motions $\mathbf{M}$ that are semantically aligned with the video content in terms of pose shape and rhythm, without requiring explicit camera parameter estimation.

2.1 Technical Framework Overview

Input Processing Pipeline

- **2D Pose Extraction:** Using pose estimators like AlphaPose or OpenPose to extract 2D skeleton sequences
- **Input Representation:** $\mathbf{p} \in \mathbb{R}^{S \times J_{2d} \times C_{in}}$ where:
 - $S = 150$ (sequence length, 5 seconds at 30 FPS)
 - $J_{2d} = 17$ (COCO format joints)
 - $C_{in} = 3$ (x, y coordinates + confidence score)
- **Preprocessing:** Missing/low-confidence joints are filtered, interpolation applied for continuity

Output Motion Representation

The target 3D motion $\mathbf{M}$ is represented per frame as:

$$\mathbf{m} \in \mathbb{R}^{S \times (24 \times 6 + 4 + 3)} \quad (12.1)$$

Breaking down the 151-dimensional vector per frame:

- **Joint Rotations (144D):** 24 joints in 6D rotation representation
- **Foot Contact Labels (4D):** Binary indicators for [left foot, right foot, left toe, right toe]
- **Root Position (3D):** Global (x, y, z) translation of pelvis

2.2 Diffusion Model Foundation

Forward Noising Process

The diffusion process follows a Markov chain, progressively adding Gaussian noise:

$$q(m_t^{1:S} | m_{t-1}^{1:S}) = \mathcal{N}(\sqrt{\alpha_t} m_{t-1}^{1:S}, (1 - \alpha_t)I) \quad (12.2)$$

The closed-form expression for any step t given clean data m_0 is:

$$m_t = \sqrt{\bar{\alpha}_t} m_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (12.3)$$

where $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

Reverse Denoising Process

The denoising network directly predicts the clean motion:

$$\hat{m}_0 = D(m_t, t, p) \quad (12.4)$$

The training objective uses the simple loss:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{m \sim q(\mathbf{M}|p), t \sim [1, N]} [\|m_0 - D(m_t, t, p)\|_2^2] \quad (12.5)$$

Analytical Posterior Distribution

The true posterior $q(m_{t-1} | m_t, m_0)$ is Gaussian with known parameters:

$$\hat{\mu}_t(m_t, m_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t}m_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}m_t \quad (12.6)$$

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t, \quad \text{where } \beta_t = 1 - \alpha_t \quad (12.7)$$

2.3 Network Architecture Details

Denoiser Network $D(m_t, t, p)$

- **Input:** Noisy motion m_t , diffusion timestep t , 2D pose sequence p
- **Architecture:** Transformer-based with three denoising modules
- **Components:**
 - Self-attention blocks for temporal coherence
 - FiLM conditioning blocks for feature-wise modulation
 - Cross-attention and MLP layers

FiLM Conditioning Mechanism

Feature-wise Linear Modulation dynamically scales and shifts features based on conditioning:

$$\mathbf{h}' = \gamma(\mathbf{c}) \odot \mathbf{h} + \beta(\mathbf{c}) \quad (12.8)$$

where:

- $\mathbf{h}$: Intermediate features of shape $n \times d$ (n frames, d hidden dimension)
- $\mathbf{c} = [\text{embed}(t); \text{encode}(p)]$: Concatenated timestep and pose embeddings
- γ, β : Functions mapping condition $\mathbf{c}$ to modulation parameters

Classifier-Free Guidance

During training, conditioning is randomly dropped 25% of the time. During sampling:

$$\hat{m}_0 = (1 + w)D(m_t, t, p) - wD(m_t, t, \phi) \quad (12.9)$$

where w controls guidance strength, balancing fidelity and diversity.

2.4 Motion Representation Details

6D Rotation Representation

Instead of problematic Euler angles or constrained quaternions, ViMo uses the continuous 6D representation:

$$\mathbf{R} = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_1 \times \mathbf{a}_2] \quad (12.10)$$

where $\mathbf{a}_1, \mathbf{a}_2$ are the first two columns of the rotation matrix. This ensures:

- Continuity for neural network learning
- No gimbal lock issues
- Easy reconstruction of orthonormal rotation matrix

Foot Contact Label Computation

Foot contact labels are computed self-supervised from foot velocities:

$$f_i = \mathbb{I}(\|\mathbf{V}_{\text{foot}}\|_2 < \epsilon) \quad (12.11)$$

where $\mathbf{V}_{\text{foot}}$ is the velocity of foot joints and ϵ is a threshold.

2.5 Loss Functions Formulation

The complete training objective combines diffusion and auxiliary losses:

$$\mathcal{L} = \mathcal{L}_{\text{simple}} + \lambda_1 \mathcal{L}_{\text{joints}} + \lambda_2 \mathcal{L}_{\text{vel}} + \lambda_3 \mathcal{L}_{\text{foot}} \quad (12.12)$$

Joints Loss (3D Position Error)

$$\mathcal{L}_{\text{joints}} = \frac{1}{S} \sum_{i=1}^S \left\| FK(R_0^i) - FK(\hat{R}_0^i) \right\|_2^2 \quad (12.13)$$

The Forward Kinematics function $FK(\cdot)$ converts joint rotations to 3D positions:

$$\mathbf{p}_j = \mathbf{p}_{\text{par}(j)} + \mathbf{R}_{\text{par}(j)} \cdot \mathbf{b}_j \quad (12.14)$$

where $\mathbf{b}_j$ is the bone vector in parent's local coordinates.

Velocity Loss (Temporal Smoothness)

$$\mathcal{L}_{\text{vel}} = \frac{1}{S-1} \sum_{i=1}^{S-1} \left\| (R_0^{i+1} - R_0^i) - (\hat{R}_0^{i+1} - \hat{R}_0^i) \right\|_2^2 \quad (12.15)$$

This ensures correct angular velocity and eliminates jitter by enforcing smooth transitions.

Foot Contact Loss (Physical Plausibility)

$$\mathcal{L}_{\text{foot}} = \frac{1}{S-1} \sum_{i=1}^{S-1} \left\| \left(FK \left(\hat{R}_0^{i+1} \right) - FK \left(\hat{R}_0^i \right) \right) \cdot \hat{f}_i \right\|_2^2 \quad (12.16)$$

Key aspects:

- Uses model’s own predicted contact labels $\hat{f}_i$
- Masks velocities of joints predicted to be in contact
- Creates self-supervised learning of foot-ground interaction

The masking operation:

$$\mathbf{V}_{\text{masked}}^i = \mathbf{m}_i \odot \mathbf{V}^i \quad (12.17)$$

where $\mathbf{m}_i$ is the contact mask broadcast across spatial dimensions.

2.6 Training and Sampling Procedure

Training Details

- **Sequence Length:** 150 frames (5 seconds at 30 FPS)
- **Diffusion Steps:** $T = 1000$
- **Optimizer:** ADAN with learning rate 1×10^{-4}
- **Sampling Acceleration:** DDIM with 50 steps
- **Conditioning:** Multi-view 2D poses from different camera angles

Sampling Process

At each denoising step:

1. Network predicts $\hat{m}_0 = D(m_t, t, p)$
2. Compute posterior mean using analytical formula:

$$\hat{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \hat{m}_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} m_t \quad (12.18)$$

3. Sample $m_{t-1} \sim \mathcal{N}(\hat{\mu}_t, \tilde{\beta}_t I)$
4. Repeat until $t = 0$ for final motion $\hat{m}_0$

2.7 Advanced Applications Implementation

Zero-Convolution for Few-Shot Stylization

For adapting pre-trained models to new styles without catastrophic forgetting:

$$W_{\text{effective}} = W_{\text{pretrained}} + W_{\text{new}} \quad (12.19)$$

where W_{new} is initialized to zeros, ensuring:

- Initial behavior identical to pre-trained model
- Gradual adaptation to target style
- Preservation of music-dance alignment knowledge

Architecture uses residual connection:

$$\text{Output} = f_{\text{frozen}}(\text{input}) + f_{\text{zero-conv}}(\text{input}) \quad (12.20)$$

Video-Guided Motion Completion

For motion inpainting and editing tasks:

$$\hat{\mathbf{m}}_{t-1} = \mathbf{b} \odot \mathbf{q}(c_{\text{const}}, t-1) + (1 - \mathbf{b}) \odot \hat{\mathbf{m}}_{t-1} \quad (12.21)$$

where:

- $\mathbf{b}$: Binary mask indicating known regions
- c_{const} : Constraint motion provided by user
- Enables style transfer and missing motion completion

2.8 Key Technical Innovations

Robustness to Camera Complexity

- No explicit camera parameter estimation required
- Multi-view 2D pose conditioning provides view invariance
- Diffusion process handles ambiguity in 2D-to-3D lifting

Self-Supervised Physical Constraints

- Foot contact labels learned jointly with motion generation
- No manual annotation or fixed thresholds required
- Model discovers physically plausible contact patterns

Structured Motion Representation

- Combined rotation, translation, and contact information
- 6D rotation ensures continuity and avoids singularities
- Hierarchical loss functions for different motion aspects

This comprehensive methodology enables ViMo to handle the challenges of casual videos while generating diverse, physically plausible 3D motions that maintain semantic alignment with input video content.

2.9 Applications

Dataset Construction

ViMo generates large-scale motion datasets from video collections, demonstrated with Chinese classic dance dataset (750 motions from 52 videos).

Few-Shot Dancing Stylization

- Uses **zero-convolution blocks** for parameter-efficient fine-tuning
- Adapts pre-trained EDGE model to new styles
- Mathematical formulation: $W_{\text{effective}} = W_{\text{pretrained}} + W_{\text{new}}$ with W_{new} initialized to zeros

Video-Guided Motion Completion

Implements motion inpainting using masked denoising:

$$\hat{m}_{t-1} = b \odot q(c_{\text{const}}, t-1) + (1-b) \odot \hat{m}_{t-1} \quad (12.22)$$

where b is a binary mask for known regions.

3 Findings and Discussions

3.1 Quantitative Results

- **Superior to MotionBERT:** Lower FID scores (15.70 vs 19.22) and higher diversity
- **User Study:** 76.05% win rate vs MotionBERT’s 23.95%
- **Stylization:** Zero-convolution approach outperforms direct fine-tuning in user preference (56.27-65.89% win rates)

3.2 Qualitative Advantages

- Handles complex camera movements and occlusions better than traditional methods
- Generates smoother, more plausible motions without jittering
- Successfully transfers styles between different dance genres

3.3 Key Insights

- **Robustness:** Works on challenging casual videos where MotionBERT fails
- **Scalability:** Can amplify limited video resources into large motion datasets
- **Flexibility:** Enables three important applications beyond basic generation

4 Conclusion

ViMo represents a significant advancement in video-to-motion generation by:

- Leveraging abundant casual video resources beyond constrained MoCap data
- Introducing a robust diffusion-based framework that handles camera complexity
- Demonstrating practical applications in dataset construction, stylization, and motion completion
- Opening new directions for semantic-aware, scene-aware motion generation

The work successfully bridges the gap between casual video content and high-quality 3D motion generation, with potential impacts across animation, gaming, and virtual reality applications.

4.1 Improvements Made

- **Novel Problem Formulation:** First to address motion generation from casual videos with complex camera movements
- **Diffusion-based Approach:** Avoids explicit camera estimation, handles ambiguity better than deterministic methods
- **Self-supervised Foot Contact:** Eliminates need for manual labeling while preventing foot sliding
- **Multi-application Framework:** Demonstrates practical utility beyond academic benchmarks

4.2 Limitations and Future Work

- **Multi-person Scenes:** Currently limited to single-person videos
- **Scene Awareness:** No interaction with environmental objects
- **Long-term Dependencies:** Limited to 5-second sequences (150 frames)
- **Evaluation Metrics:** FID and diversity metrics may not fully capture perceptual quality